

Intro Network Security

L00: Course Overview and Logistics

Dr. Yiwen Hu

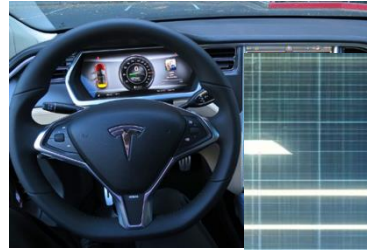
CMSC 681/487 — Intro Network Security



Pacemaker & Monitor



Laptop



cars



Slingbox: remote control cable



Security Camera



AR devices



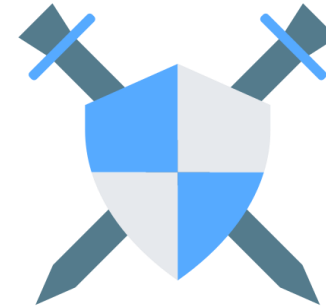
Internet phones



Gaming devices



Fitbit



Security Job Market

Security Analyst jobs:

- 182,800 (2024) → 234,900 (2034)
- +29% growth — much faster than average
- Source: U.S. Bureau of Labor Statistics (BLS), Occupational Outlook Handbook

Big-Company Demand Signals

- Top hiring sectors:
 - Cloud providers (AWS, Azure, Google Cloud), Apple, Tesla, SpaceX, government agencies
- Security-focused firms expanding:
 - Cisco, Palo Alto Networks, Fortinet, CrowdStrike

Security roles grew even during tech layoffs

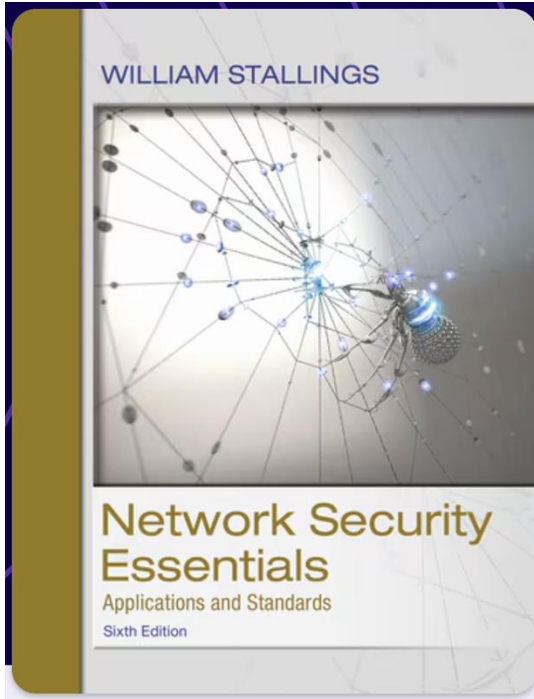
Course Information

- **Credit units:** 3
- **Location:** Information Technology (ITE) 104
- **Date:** 08/24/2026 - 12/09/2026
- **Lecture time:** Monday and Wednesday 2:30PM - 3:45PM
- **Prerequisite:**
 - Undergraduate: CMSC 421 or CMSC 481 (grade of C or better)
 - Graduate: CMSC 341 and CMSC 481

Office Hour

- **Instructor:** Dr. Yiwen Hu <huyiwen@umbc.edu>
 - **Location (In-Person):** ITE 322-A
 - **Time:** Monday, 3:45 PM – 4:45 PM, or by Appointment
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- **TA:** Haitian Yan <hyan1@umbc.edu>
 - **Location (In-Person):** ITE 352
 - **Webex Link (Online):** <https://umbc-students.webex.com/meet/hyan1>
 - **Time:** Thursday, 11:00 AM -12:00 PM, or by Appointment

Textbook and Reading Materials

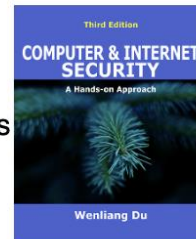


You can find it on Blackboard

**Computer & Internet Security:
A Hands-on Approach**
3rd Edition
(ISBN: 978-17330039-4-0)

Topics

- Software security
- Web security
- Meltdown/Spectre attacks
- Cryptography, PKI
- Selected topics from network security



Textbook: recommended,
not required

Additional readings: may
be shared in lectures to
spark interest & deepen
understanding

<https://www.handsonsecurity.net/>

Communications

- **Blackboard:** announcements, course materials, submissions
- **Discord:** general questions/clarifications (the link will be provided on 08/26)
 - Please don't post solutions or share materials that directly reveal answers
- **Email:**
 - Start your subject line with **[CMSC487/687]**
 - Send to instructor or/and TA via your UMBC email
 - We mostly reply within 24 hrs (weekdays)
 - Last-minute or non-compliant emails may not get a response

Grading Breakdown

Component	Weight	Ind./Group	Description
Exam	25%	Individual	One exam, held before Thanksgiving.
Homework assignments	20%	Individual	Three assignments consisting of conceptual questions or security tool implementations.
Course project	35%	Group (3–4)	Project proposal (5%), final presentation (10%, about 15 minutes of presentation and Q&A), and final report and deliverables (20%); graded at both the group and individual level.
In-class quizzes	10%	Individual	Three to four short quizzes with conceptual questions or exercises.
Research paper presentation & discussion	10%	Group (3–4)	Each group selects one paper from an assigned list for a 15-minute in-class presentation and discussion.
Total	100%		

Grading Scale

Grading scale: Given a percentage X of possible points:

- $90\% \leq X$ is A,
- $80\% \leq X < 90\%$ is B,
- $70\% \leq X < 80\%$ is C,
- $60\% \leq X < 70\%$ is D,
- $X < 60\%$ is F

Depending on class performance, **it is possible the grade cutoffs will be adjusted lower than these thresholds.** They will not be higher.

The Important Stuff – Only One Exam

- Exam: in class, in October
- Details regarding the exact exam date and permitted materials will be announced in a later class
- No makeup exams except in extreme circumstances with valid, documented proof of absence

The Important Stuff - Homework Assignments

- Homework: 3 assignments (concepts, exercises, or network tools)
 - Two assignments before the midterm exam; One assignment after the midterm exam
 - The tentative schedule for all homework assignments is provided in the course syllabus

All homework assignments are individual work. They won't take up too much time. Collaboration, copying, or sharing solutions is not permitted

Due dates (typically 7–14 days after release) and submission instructions for each assignment will be announced on Blackboard, with a standard deadline of 11:59 PM Eastern Time (UMBC's local time zone) unless stated otherwise.

The Important Stuff – One Course Project

- Each student will participate in one group project
 - 3-4 students per group, about 10 groups
- Topics must relate to network security
 - The instructor will suggest some project topics. Groups are also welcome to propose their own project ideas.
 - Please do not propose projects that require hardware your team does not currently have or cannot reliably access
 - Students who assist teams in addressing hardware needs will receive bonus points
 - For promising projects, the instructor can help to find equipment you may need

The Important Stuff – One Course Project

- **Please form your group and discuss your interests as early as possible** to make the project run smoothly. Once your group is ready:
 - Pick a group leader
 - Have the group leader email instructor with:
 - Your group name (**feel free to be creative** 😊)
 - All group members CC'd on the email so everyone is included

The Important Stuff – One Course Project

- Everyone is expected to actively contribute at each stage of the project
- **Make sure your individual contributions are clear in project submissions**—they count toward your grade!
- **The final presentation will be graded by both the instructor and your classmates**
 - About 13 minutes presentation + 2 minutes Q&A and grading
 - Feel free to be creative and make your presentation engaging
- You'll also complete an **anonymous peer evaluation** to give feedback on your group members, which will help determine individual grades

The Important Stuff – One Course Project

- The project (35%) will be graded as follows:
 - 5% for the project proposal (Due: around Oct.14)
 - 3–5 pages, including references, single-column format. A template will be provided
 - 10% for the final presentation (Dec. 2 and Dec.7, 13-min presentation + 2-min Q&A)
 - 20% for the final report and validation of submitted project code/materials
 - Must follow the IEEE CNS conference paper format and be 6–9 pages long (double-column), including references

Course Project Rules

- **IRB Consideration:** The proposed project must not pose risks to any individuals, animals, or public/production systems, including (but not limited to) the UMBC email system, UMBC Wi-Fi, or other public networks or services. Large-scale attacks are strictly prohibited (*More details: <https://research.umbc.edu/federalmaryland-regulations-and-policies/>*)
- **Project Approval Requirement:** The project may proceed only after this proposal has been approved by the instructors. The project will be conducted during the Fall 2026 semester only. If, at any point during the project period, the group is uncertain whether any proposed steps are permissible, the group must consult with the Instructor or a TA before proceeding

Failure to comply with these requirements may result in failure of the course and additional disciplinary action as deemed appropriate by the university

The Important Stuff - Homework Assignments and Project

Late assignments and project will be accepted with the following penalties:
applied to the earned score:

- Up to 12 hours late: -10%
- 12–24 hours late: -20%
- 24–48 hours late: -40%
- More than 48 hours late: no credit

The Important Stuff - Quizzes

- Four in-class quizzes
- 3–5 questions: true/false, multiple-choice, or short answer.
- **No makeup quizzes**

Quiz Notice Policy: If attendance stays strong, I'll give advance notice of quiz dates. If it drops too low, I won't — showing up should come with a reward.

The Important Stuff –Research Paper Presentation & Discussion

- November lectures cover advanced network security topics delivered by the instructor, with up to two student research paper presentations integrated into each session
 - A list of research papers will be provided for groups to select from. The papers are closely related to the topics covered in the corresponding lectures
 - Each group (3-4 students) selects one paper from an assigned list for a 15-minute in-class presentation and discussion

Academic Integrity

This course adheres to the academic honesty policy outlined at <https://academicconduct.umbc.edu/>

The following is a brief summary of the policy:

Academic integrity is an important value at UMBC. By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC's scholarly community in which everyone's academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal.

Academic Integrity -Use of AI Tools

Exams and Quizzes:

- The use of AI tools, including, but not limited to, language models such as ChatGPT, is **strictly prohibited for exams and quizzes**. There are no exceptions

Academic Integrity -Use of AI Tools

Homework Assignments, Course Projects, and Research Paper Presentation & Discussion:

- AI tools may be used only to support your understanding (e.g., clarifying concepts), and **any use of AI must be proactively disclosed** in your submission
- **You are strictly prohibited from using AI** to directly complete your work, such as, but not limited to, completing assignments, generating project topics, writing reports, preparing presentation slides, or writing code
- All submissions will be checked rigorously; any prohibited or undisclosed use of AI constitutes academic dishonesty and will be handled under the academic integrity policy

- **Reminder:** Please do not post any course materials online or share them with others without the instructor's permission

UMBC Policies

- Accessibility and Disability Accommodations, Guidance and Resources
- Sexual Assault, Sexual Harassment, Gender Based Violence and Discrimination
- Pregnant and Parenting Students
- Religious Observances & Accommodations
- Hate, Bias, Discrimination and Harassment

Lecture Organization (Tentative)

Lecture	Date	Subject
Part I: Fundamentals		
L01	Aug. 24, 2026	Course Overview, Introduction to Network Security
L02	Aug. 26, 2026	Networking Fundamentals Review
L03	Aug. 31, 2026	Cryptographic Tools for Confidentiality and Integrity
L04	Sep. 2, 2026	Symmetric Encryption (HW1 Release)
	Sep. 7, 2026	Labor Day, no lecture
L05	Sep. 9, 2026	Symmetric Encryption
L06	Sep. 14, 2026	Public Key Cryptography (HW1 Due)
L07	Sep. 16, 2026	Public Key Cryptography
L08	Sep. 21, 2026	User Authentication
L09	Sep. 23, 2026	User Authentication (HW2 Release)
L10	Sep. 28, 2026	Network Access Control
L11	Sep. 30, 2026	Network Access Control
L12	Oct. 5, 2026	Transport-Layer Security (HW2 Due)
L13	Oct. 7, 2026	Transport-Layer Security
	Oct. 12, 2026	Fall Recess, no lecture
L14	Oct. 14, 2026	IP Security (Project Proposal Due)
L15	Oct. 19, 2026	Exam Review
L16	Oct. 21, 2026	Exam

**May be adjusted
depending on how the
class progresses**

Lecture Organization (Tentative)

Part II: Advanced Topics and Research Frontiers		
L17	Oct. 26, 2026	Passive and Active Attacks
L18	Oct. 28, 2026	Widely-Used Network Security Tools (HW3 Release)
L19*	Nov. 2, 2026	Wireless Network Security (Wi-Fi)
L20*	Nov. 4, 2026	Wireless Network Security (IoT, V2X)
L21*	Nov. 9, 2026	Cellular Network Security (5G) (HW3 Due)
L22*	Nov. 11, 2026	Cellular Network Security (NTN)
L23*	Nov. 16, 2026	Cellular Network Security (Emerging applications and services)
L24*	Nov. 18, 2026	Machine Learning for Network Security
L25*	Nov. 23, 2026	Adversarial Machine Learning and Security of AI Systems
	Nov. 25, 2026	Thanksgiving Break, no lecture
L26*	Nov. 30, 2026	Security Risks of Agentic Network Operations
L27	Dec. 2, 2026	Project Final Presentations
L28	Dec. 7, 2026	Project Final Presentations
L29	Dec. 9, 2026	Project Report Writing Instructions, Class Summary

**May be adjusted
depending on how the
class progresses**

* November lectures cover advanced network security topics delivered by the instructor, with up to two student research paper presentations integrated into each session. A list of research papers will be provided for groups to select from. The papers are closely related to the topics covered in the corresponding lectures.